

Editorial

Software process improvement leveraged in various application domains

1. INTRODUCTION

The European System and Software Process Improvement and Innovation Initiative (EuroSPI²) was founded in 1994. EuroSPI²'s mission is to develop an experience and knowledge exchange platform for Europe where SPI practices can be discussed and exchanged, and knowledge can be gathered and shared. This mission is implemented by the following major action lines (<http://www.eurospi.net>):

- An annual EuroSPI² conference supported by Software Process Improvement Networks from different European countries.
- EuroSPI² supported the establishment of a worldwide SPI Manifesto with SPI values and principles agreed among experts worldwide. We build clusters of experts and knowledge libraries for these values and principles.
- Establishing an Internet-based knowledge library based on hundreds of experience reports has been contributed to EuroSPI² since 1994.
- Establishing a European Qualification Framework for a pool of professions related to SPI and management. This is supported by European certificates, examination systems, and online training platforms (European Certification and Qualification Association).
- Establishing a worldwide newsletter with articles from key industry and key European research associations helping to implement the SPI Manifesto worldwide.

A typical characterization of EuroSPI was stated by a company using the following words: ‘... the biggest value of EuroSPI lies in its function as a European knowledge and experience exchange mechanism for SPI and innovation’.

EuroSPI²2012 was held for the second time in Vienna, the capital of Austria.

In 2012, EuroSPI² continued with the open workshop communities aligned with the conference and focusing on specific SPI topics. This included Creating Environments Supporting Innovation and Improvement, SPI and Product, System, Software Design, and SPICE Assessors: Exchanging Experiences Across Assessment Models, Business Process Innovation and Improvement, SPI and Measurement, and SPI in small-sized and medium-sized enterprises.

Also, at EuroSPI²2012, we continued to extend the scope of the conference from software process improvement to systems, software and services process improvement.

Among the 26 countries represented at EuroSPI²2012, the contributions were highly relevant for the European and worldwide industry also appearing in this special issue. The SPI Manifesto from 2009 created a future vision, the extension of SPI to the systems and product level led to the involvement of major European manufacturing firms, the building of workshop communities for specific topics created further networking opportunities, and the contributions included good practices about how to achieve that.

Systems, software and services process improvement is meanwhile a major learning and success factor worldwide for staying competitive on a global and dynamic developing world market.

2. EUROSPI BOARD

Since 1994, the EuroSPI initiative is financed and coordinated by a group of leading applied research companies and quality networks. Since 1998, the following board members have been important drivers for the initiative:

American Society for Quality Software Division, <http://www.asq.org>.

Association for Software Quality and Further Education, <http://www.asqf.de>.

DELTA, Denmark, <http://www.delta.dk>.

International Software Consulting Network, Ireland and Austria, <http://www.iscn.com>.

SINTEF, Norway, <http://www.sintef.no> and.

Software Technology Transfer Finland and Finish Software Measurement Association., <http://www.fisma.fi>.

In 2012, the Institute of Software Technology and Interactive Systems of Vienna University of Technology and the International Network for Terminology (TermNet) collaborated with this board to coordinate with EuroSPI²2012.

3. ABOUT THE SPECIAL ISSUE

The value of this selection of papers lies in the fact that they present actual industrial experiences. Research-oriented EuroSPI²2012 papers were published in separate proceedings.

Since its beginning in 1994 in Dublin, the EuroSPI² initiative outlines that there is no single silver bullet to solve SPI issues, but you need to understand a combination of different SPI methods and approaches to achieve real benefits. Therefore, each proceeding covers a variety of different topics and, at the conference, we discuss potential synergies and the combined use of such methods and approaches.

This proceeding contains a selection of industrial experience papers leveraging SPI in various application domains, which reflect EuroSPI²'s multidimensional approach with the three fundamental components of process improvement known to be people and skills, process, and technology.

The paper by Phelim Dowling describes the transformation of a research project into a fully fledged commercial entity with an emphasis on the software process and quality methodologies used that in this case are mainly based on AGILE methods. In this paper, the author outlines how to engage in the software process, specifically AGILE, from an early stage transforming a research project into a commercial spinout.

The paper 'Methodological framework for the allocation of work packages in GSD' by Ruano-Mayoral, Casado-Lumbreras, Garbarino-Alberti, and Misra presents and tests a framework devoted to allocate tasks among project participants in global software development settings. This framework is tested in real projects with promising results in terms of lower defects introduction and increased overall satisfaction from all parties.

Onur Kaynak and N. Alpay Karagöz present an experience report about the integrated management of the ISO (International Organization for Standardization)/IEC (International Electrotechnical Commission) 15504, ISO/IEC 27001, ISO/IEC 20000, and ISO/IEC 9001:2008 in an information technology solutions provider company. This paper depicts the process improvement initiative challenges with the four different management systems.

The paper by Hervé Cholez and Frédéric Girard proposes a method that is aimed to conduct a first assessment of enterprise information security maturity and consequently improves its process accordingly. The research work has been performed by the Public Research Centre Henri Tudor, Luxembourg in collaboration with the Luxembourg's Ministry of Economy and Foreign Trade and is specifically designed for small-sized and medium-sized enterprises.

The paper by Martin McHugh, Fergal McCaffery, and Valentine Casey presents the results of a survey devoted to study the barriers on the adoption of AGILE practices in the medical device industry. Medical device developers must develop software in accordance with regulatory requirements and as a result, a number of the barriers to AGILE adoption are associated with the process of achieving regulatory conformance.

Tomas Schweigert *et al.* present a study devoted to investigate AGILE Maturity Models by means of an industrial survey. Results show that we cannot find a common accepted AGILE Maturity Model, and extensive research on the area is needed to develop a common proposal from the information available.

Hideto Ogasawara, Takumi Kusanagi, and Minoru Aizawa propose an SPI framework and introduce the results of SPI activities based on Toshiba's approach that has been implemented since 2000.

ACKNOWLEDGEMENTS

The editors would like to express their thanks to the members of the EuroSPI²2012 Industrial Programme Committee for their contribution to the review and selection of the papers:

Bachmann Volker, SIBAC GmbH, Germany; Bæk Jørgensen Jens, Mjølner Informatics A/S, Denmark; Breske Eva, Robert Bosch GmbH, Germany; Christiansen Mads, DELTA, Denmark; Daughtrey Taz, James Madison University, USA; Dekkers Carol, Quality Plus Technologies, USA; Dussa-Zieger Klaudia, Method Park Software AG, Germany; Ekert Damjan, International Software Consulting Network Ltd GmbH, Austria; Fehrer Detlef, SICK AG, Germany; Fenz Andrea, Skills International GmbH, Austria; Forselius Pekka, 4SUM Partners, Finland; Georicke Stephan, International Software Quality Institute GmbH, Germany; Hagenmeyer Philipp, ZF Friedrichshafen AG, Germany; Hallikas Jarmo, Falcon Leader Oy, Finland; Hällmayer Frank, Software Factory GmbH, Germany; Hind Tim, AXA, UK; İbrişim Ayşegül, Turkish Standards Institution, Turkey; Johansen Jørn, DELTA Axiom, Denmark; Johansson Mika, Finish Software Measurement Association, Finland; König Frank, ZF Friedrichshafen AG, Germany; Messnarz Richard, ISCN Ltd, Austria/Ireland; Morgenstern Jens, Germany; Nevalainen Risto, FiSMA, Finland; O'Leary Eugene, EQN Ltd., Ireland; Poth Alexander, Software Quality Systems AG, Germany; Renault Samuel, Centre de Recherche Public Henri Tudor, Luxembourg; Romcea Cristina, Conti Temic microelectronic GmbH, Germany; Sauberer Gabriele, TermNet, Austria; Schweigert Tomas, SQS AG, Germany; Sechser Bernhard, Method Park, Germany; Spork Gunther, Magna Powertrain, Austria; Stefanova-Pavlova Maria, Center for Innovation and Technology Transfer-Global, Bulgaria; Von Bronk Peter, Systemberatung Software-Qualität, Germany; and Wegner Thomas, ZF Friedrichshafen AG, Germany.

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